

$$L = (0.35 \pm 0.05) \text{ m}$$

$$T_1 = (32.0 \pm 0.5) ^\circ\text{C}$$

$$T_2 = (4.0 \pm 0.5) ^\circ\text{C}$$

What is the fractional uncertainty in R ?

A 0.11

B 0.14

C 0.16

D 0.18

Ans: D

$$\begin{aligned}\frac{\Delta R}{R} &= \frac{\Delta L}{L} + \left| \frac{\Delta(T_2 - T_1)}{(T_2 - T_1)} \right| \\ &= \frac{0.05}{0.35} + \left| \frac{\Delta T_2 + \Delta T_1}{(T_2 - T_1)} \right| \\ &= \frac{0.05}{0.35} + \left| \frac{0.5 + 0.5}{(4.0 - 32.0)} \right| = 0.18\end{aligned}$$

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A value for the acceleration of free fall on Earth is given as $(10 \pm 2) \text{ m s}^{-2}$